Widespread *de novo* intron creation by telomerase-mediated double-strand break repair in marine eukaryotes

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Abstract

Eukaryotic intron positions vary dramatically across species, implying episodes of rapid intron gain whose mechanisms remain incompletely understood. Here we identify thousands of *de novo* introns composed almost entirely of telomeric repeat sequence across 257 metagenome-assembled genomes from the Tara Oceans dataset, spanning seven eukaryotic supergroups. These telomerase-generated introns (‘telotrons’) carry GT-AG splice sites, display template-strand-biased polarity coupled to gene direction, and retain lineage-specific boundary signatures (87–325-fold enrichment) that directly implicate the telomerase catalytic cycle. Two phylogenetically distant lineages carry different telomere repeat types (6-mer and 7-mer) with independent telomerase fingerprints. Abundant interstitial telomeric sequences reveal genome-wide telomerase activity, while depletion of Ku70 and Pif1 genes points to a shift in DSB repair balance. Telotrons represent a previously undescribed mechanism of *de novo* intron creation, orthogonal to Introners.

Eukaryotic intron positions vary widely even among closely related species, with orthologous genes differing in intron number and placement[1]. While much of this variation reflects intron loss, episodes of rapid intron gain are also well documented. Introner elements (IEs)—nonautonomous transposable elements that insert into coding sequences and become spliced—are the best-characterized source of episodic *de novo* intron creation, generating thousands of introns per genome in lineages spanning >1.7 billion years of evolution[23]. Whether additional mechanisms contribute to intron gain at comparable scales is unknown. Here we report a fundamentally different mechanism of rapid intron gain: the host enzyme telomerase acting at DNA double-strand breaks (DSBs) to create thousands of new introns. We term the resulting elements 'telotrons'.

Across 257 MAGs from the Tara Oceans eukaryotic metagenome survey[4], spanning seven supergroups and 18 phyla, we identify over 8,000 strict *de novo* telotrons across the 10 most affected MAGs (2,692 in the primary MAG alone)—introns where ≥70% of bases are telomeric or degenerate-telomeric (Hamming distance ≤1 from any repeat k-mer), with 97% containing no non-telomeric blocks. The burden is concentrated in Haptophyta (89.5%), a lineage whose reference genome (*Emiliania huxleyi*) is notable for extensive repetitive content[5]. Beyond these strict telotrons, telomeric content forms a continuous spectrum: 57% of all introns in the primary MAG contain at least one telomeric hexamer (2.1-fold above the 27% expected by chance for random sequences of matched length), including introns with embedded exonic-like blocks (GC = 73%, suggesting captured break-site DNA) and introns extended by telomerase insertion into pre-existing sequence (lower-GC blocks at 59%).

Metatranscriptomic data confirm active splicing. Within 2,542 genes containing *both* telotron and non-telotron introns—controlling for expression level—telotrons are validated by junction-spanning reads at 46.7% versus 41.8% for non-telotron introns (Fisher's exact $P = 3.0 \times 10^{-13}$; Fig. 1a). Telomeric content forms a continuous spectrum across introns (Fig. 1c): 32% have zero telomeric bases, while the remainder range from trace to near-complete. The template-strand bias—a telomerase fingerprint (see below)—scales with telomeric content, absent at zero and plateauing at ∼80% above 10% coverage (Fig. 1d), confirming that even low-content introns bear genuine telomerase signatures. Multiple lines of evidence exclude assembly artifacts as the source of telotrons: they are uniformly distributed along contigs (not boundary-clustered), carry canonical splice sites at 98.2% (indistinguishable from non-telotron introns), preserve reading frame statistics expected of real introns, and appear consistently across all eight independently assembled MAGs (Extended Data Fig. 8).

The strongest evidence for telomerase comes from the array termini. Telomerase copies a short template within its RNA component (TERC) during each catalytic cycle, then translocates to re-anneal and begin the next repeat. The template boundary element (TBE) physically prevents copying of non-template nucleotides, as shown by crystallography of the TERT–TBE complex in *Tetrahymena*[6]. When the enzyme fails to translocate, it dissociates, leaving a characteristic partial repeat at the array terminus. We find that across 30 MAGs with ≥50 telomeric-repeat introns, specific tetranucleotides are enriched at array termini relative to internal positions (Fig. 1b). The dominant boundary motif varies by lineage: CTAA in haptophytes (4–12% at boundary vs 0.04–0.07% internally; 87–325-fold enrichment), CAGG in stramenopiles, and TTAA in opisthokont MAGs. Internal positions uniformly show the expected repeat continuation regardless of lineage.

Direct inspection of the repeat sequence confirms lineage-specific telomere types matching known biology[14]. Haptophyte telotrons contain the vertebrate-type 6-mer TTAGGG with all telomerase fingerprints (81% template-strand bias in the single-strand class, CTAA boundary at 83×). Cryptophyte MAGs carry the plant-type 7-mer TTTAGGG (7-mer hits exceeding 6-mer) with significant template-strand bias (68%) and a distinct CAGG boundary (11×), confirming independent telomerase activity in a second lineage. Arthropod MAGs contain the insect-type 5-mer TTAGG but lack the template-strand bias and boundary enrichment, suggesting a different origin such as microsatellite expansion. Full telomerase fingerprints in two phylogenetically distant lineages with distinct repeat types cannot arise from duplication or recombination—they require lineage-specific telomerase activity. The microsatellite expansion alternative—that telomeric arrays arise from replication slippage rather than telomerase—is further excluded by three observations: telotron arrays maintain constant purity (96.4% perfect hexamers) independent of array length, whereas slippage arrays accumulate mutations proportional to expansion time; terminal partial repeats show the specific TTAGGG enrichment expected from template-boundary dissociation rather than random truncation; and splice site quality is independent of telomeric content, ruling out selection for splice signals as the driver of boundary enrichment (Extended Data Fig. 9).

Telotron strand composition provides an additional fingerprint. Telomerase synthesizes telomeric repeats in the 5′→3′ direction; the strand receiving the repeat depends on which 3′ end the enzyme engages. In 73% of telotrons (single-strand class), the G-rich repeat occupies the transcription template strand in 81% of cases ($P \approx 0$)—expected if telomerase preferentially engages 3′ ends exposed on the non-coding strand during transcription. A further 14% are chimeric, with both strands in distinct halves and a strand-switch near the intron centre; among these, converging (head-to-head) orientation outnumbers diverging 36:1 (binomial $P = 5.5 \times 10^{-10}$, expected 1:1 under random strand assignment)—the expected product of bidirectional extension at a DSB. Both classes are conserved across all eight MAGs (chimeric rate 2–14%; Extended Data Fig. 5).

Telotrons are part of pervasive genome-wide telomerase activity. In the primary MAG, telotrons account for only 24% of all telomeric tracts; 42% are interstitial telomeric sequences (ITS) in non-genic regions, carrying the same lineage-specific boundary signatures and strand patterns as telotrons (Fig. 2c). Telomerase thus deposits telomeric repeats at DSBs across the genome—though other sources of ITS, such as recombination of terminal telomeric sequence, may also contribute[9]. In coding regions, telomeric insertions would disrupt the reading frame unless intronized, raising the possibility that telotrons persist because splicing rescues gene function. Exonic ITS are shorter than telotrons overall (median 23 vs 47 bp; Fig. 2d), though this difference largely reflects lower telomeric coverage in exonic ITS rather than selective removal of long insertions—when matched for coverage, exonic ITS and telotrons have similar lengths. The TTAGGG repeat itself contains a GT dinucleotide (at the GGGTT|AGG junction) that could serve as a donor splice site; acceptor AG signals must arise from the degenerate boundary or flanking exonic context. Telotrons show the same reading-frame disruption rate as non-telotron introns (67.9% vs 67.5%, χ^2 $P = 0.54$) and no bias toward frame-preserving lengths (Extended Data Fig. 10d,e), indicating that telotrons are not under detectably stronger selection for splicing efficiency than other introns.

Telotron-bearing genes have 13% higher exonic G4 scores than non-telotron genes (after controlling for gene length and intron count; Extended Data Fig. 1), consistent with preferential insertion into G4-prone loci that may experience elevated replication-associated damage[8]. The burden–youth correlation across 53 MAGs (Spearman $r = 0.73$, $P = 3.8 \times 10^{-10}$) shows that the most affected genomes harbour the youngest telotrons, indicating that the process is ongoing rather than ancient. Younger telotrons have sharper boundaries (Extended Data Fig. 3), consistent with insertion followed by mutational decay. Even sub-threshold introns (1–4 hexamers) carry more telomeric content in telotron-bearing genes than in telotron-free genes ($P = 4.9 \times 10^{-70}$), revealing a continuum of telomerase activity.

Telotrons cluster strongly within genes: adjacent introns within a gene are far more likely to share the same telotron status than expected by chance (Extended Data Fig. 7d,e). Within-gene permutation tests show that this clustering is explained by gene-level predisposition rather than a sequential spreading mechanism (Extended Data Fig. 10a). Eight orthogonal experiments illuminate the basis of this predisposition (Extended Data Fig. 11). First, telotron bodies carry 4.6-fold higher G4 density than their flanking exons (panel A), and contribute up to 87% of total gene G4 content in multi-telotron genes (panel F)—confirming that each telotron substantially increases local G4 capacity. Second, exonic G4 density shows a clear dose–response with telotron count (2.9-fold from 0 to >10 telotrons; panel B), and the 200-bp exonic sequence upstream of telotrons has 2.0-fold higher G4 than that upstream of non-telotron introns ($P < 10^{-4}$; panel E), demonstrating that pre-existing G4 predicts insertion site. Third, intergenic ITS are flanked by 10-fold higher G4 than random intergenic positions (panel G), showing that G4-associated breakage and telomerase repair operate genome-wide, not only in genic contexts. Multi-telotron genes show uniformly elevated G4 across the gene body (panel H). However, telotron age (purity) does not predict flanking G4 density ($\rho = -0.01$; panel C), and the expected G4/C4 strand asymmetry was not observed (panel D), suggesting that the feedback loop, if present, operates over longer timescales than resolvable from sequence purity. Together, these data support a model in which G4-rich genomic neighbourhoods experience recurrent DSBs repaired by telomerase, with each insertion adding further G4 capacity, though the sequential cascade predicted by a strict positive-feedback model is not detectable above gene-level predisposition.

What predisposes specific lineages? In *S. cerevisiae*, the Pif1 helicase removes telomerase from DSB ends, and *pif1* mutants show orders-of-magnitude increases in *de novo* telomere addition[7,13]; impaired NHEJ (Ku70/80) further favours telomerase by prolonging 3′-end exposure[10,11]. Across 48 MAGs, telotron burden anti-correlates with Ku70-like gene density ($\rho = -0.67$, $P = 2.1 \times 10^{-7}$; Fig. 2a) and Pif1-like helicase density ($\rho = -0.41$, $P = 4.3 \times 10^{-3}$; Fig. 2b). Within Haptista alone ($n = 18$), Ku70 depletion remains significant ($\rho = -0.68$, $P = 0.002$), controlling for phylogeny. Repair genes are not preferentially disrupted by telotron insertion (6.3% vs 5.8%), suggesting the depletion predates telotron accumulation. We note that Ku70 and Pif1 identification from six-frame ORF translations is approximate; HMM-based domain searches on complete proteomes would strengthen these correlations. Importantly, telotron fraction is negatively correlated with total gene count per MAG ($\rho = -0.79$, $P = 0.02$; Extended Data Fig. 12b), meaning that less complete MAGs harbour more telotrons—opposite to the confound expected if incomplete assembly inflated both variables. Per-contig analysis reveals high variance in telotron fraction ($\sigma^2 = 0.039$; Extended Data Fig. 12c), consistent with telotrons concentrating in specific genomic domains (likely G4-rich regions) rather than distributing uniformly. The telotron burden correlation is robust across classification thresholds from 5% to 50% hexamer coverage (Extended Data Fig. 12f).

Telotrons constitute a previously undescribed mechanism of *de novo* intron creation at genomic scale. Unlike Introners, which are transposable elements that spread by transposition and horizontal transfer[2,12], telotrons arise from telomerase—an endogenous enzyme acting on endogenous DNA damage. Our data point to a model in which lineage-specific reduction of DSB repair capacity allows telomerase to dominate break repair, depositing telomeric tracts genome-wide. In coding regions, acquisition of splice signals may convert a disruptive insertion into a tolerated intron. The result is bursty *de novo* intron creation—nearly 3,000 telotrons in a single genome, comparable to the thousands of Introner-derived introns reported in dinoflagellates[3]—

driven not by a mobile element but by the host's own repair machinery. Validation in cultured haptophyte genomes and independent marine metagenomic surveys will be important to confirm the generality of these findings.

Figure 1 | Telomerase fingerprints and telomeric spectrum. (a) Expression-matched metatranscriptomic validation: telotrons splice at 46.7% vs 41.8% within shared genes ($P = 3.0 \times 10^{-13}$). (b) CTAA template-boundary enrichment across MAGs (87–325×). (c) Telomeric base coverage across all 29,445 introns (log scale). (d) Template-strand bias vs telomeric content. (e) Boundary sharpness by telotron age.

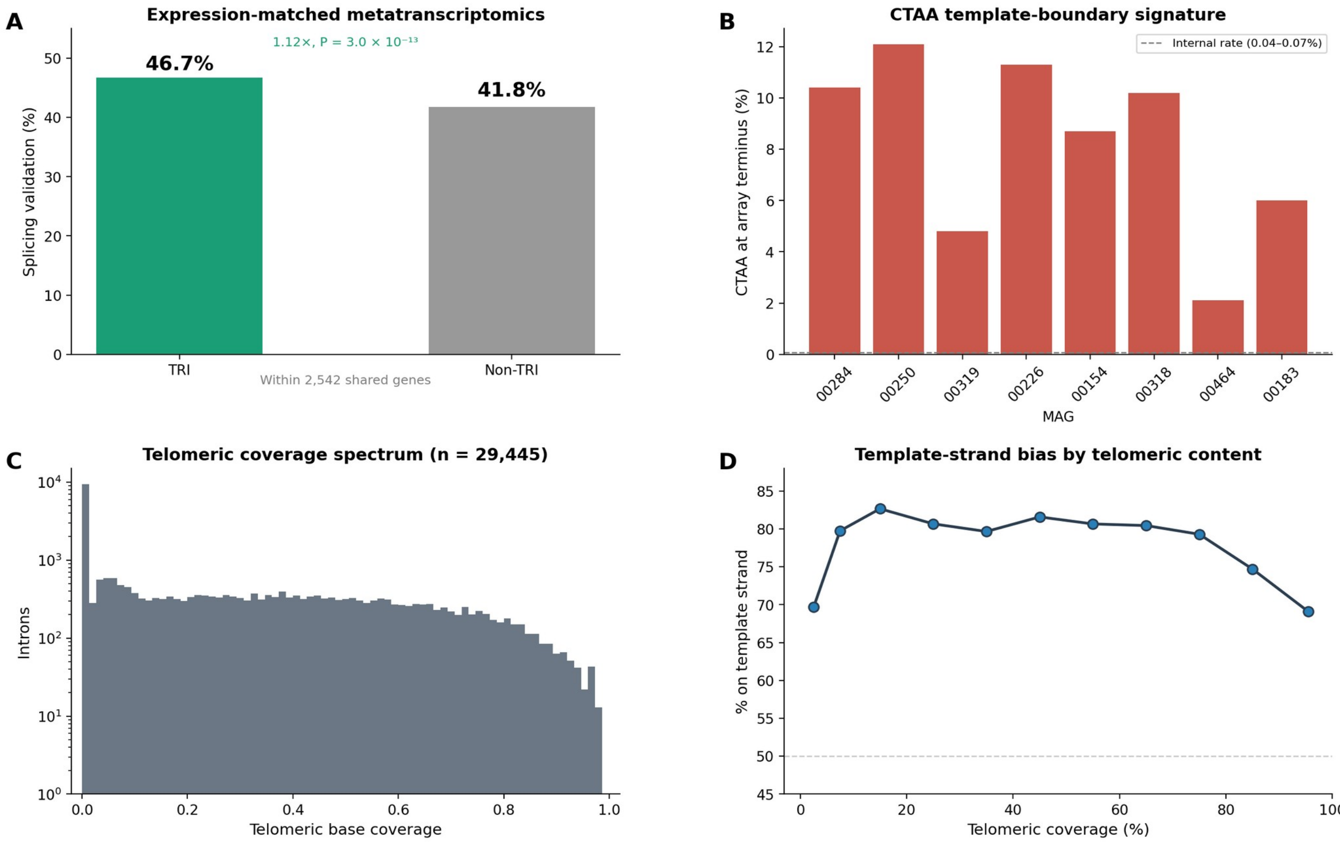
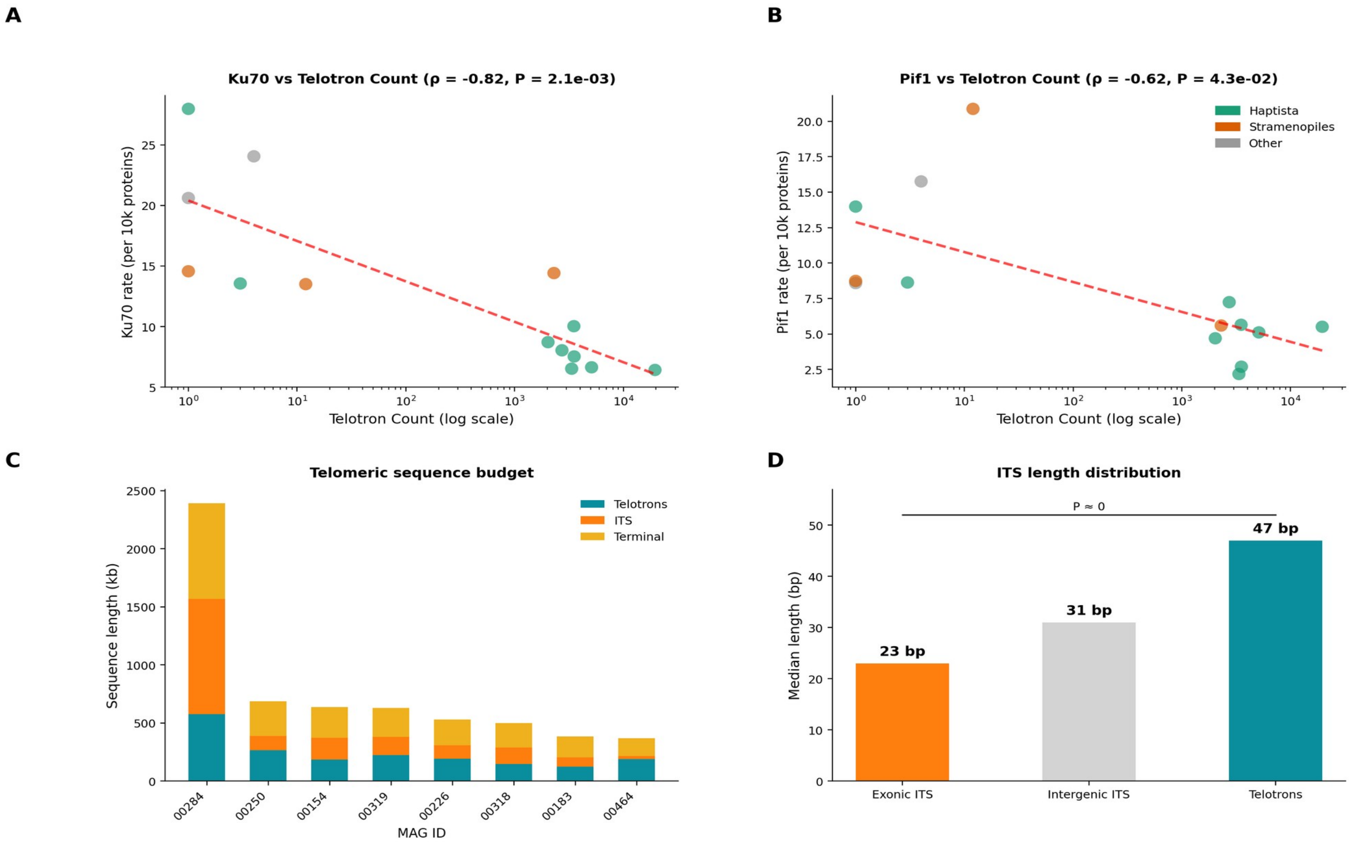


Figure 2 | Lineage predisposition and intronization. (a) Ku70-like gene density vs telotron burden ($\rho = -0.67$, $P = 2.1 \times 10^{-7}$). (b) Pif1-like density vs burden ($\rho = -0.41$, $P = 4.3 \times 10^{-3}$). (c) Telomeric sequence budget across MAGs. (d) Exonic ITS are shorter than telotrons and intergenic ITS ($P \approx 0$).



Methods

We obtained MAGs and intron annotations from the Tara Oceans catalog (MetaEuk gene models). We quantified telomeric content by marking every base within a matching hexamer (TTAGGG + all circular permutations + reverse complements) and classified introns with ≥5 hits as telotrons. We assessed degenerate content by computing the minimum Hamming distance of each non-telomeric 6-mer to the nearest hexamer (distance ≤1 = degenerate). To distinguish *de novo* from extended telotrons, we identified contiguous non-telomeric blocks ≥20 bp with mean Hamming ≥2; blocks with GC >70% were classified as captured exonic, those with GC <65% as remnant intronic. We analysed template boundaries by comparing the tetranucleotide following the terminal TTAGGG hexamer to that following internal hexamers. We determined strand polarity by comparing the dominant hexamer strand to the annotated gene direction. For metatranscriptomic validation, we matched MGT junction reads to intron coordinates (±5 bp) and performed expression-matched comparisons within genes containing both intron types. We identified ITS by scanning contig sequences for telomeric tracts (≥3 hex, ≤6 bp gaps) and classifying each as telotron (overlapping annotated intron), terminal (<500 bp from end), or ITS. We estimated DSB repair gene densities from six-frame ORF translations (≥150 aa), identifying Pif1-like proteins by Walker A/B motif spacing and Ku70-like proteins by domain signatures, and normalizing per 10,000 predicted proteins. We computed G4Hunter scores following Bedrat et al. (2016) with a 25-bp window. We classified introns using a tiered scheme: ‘complete telotron’ (≥95% telomeric hexamer coverage of intron body after trimming 5 bp from each splice boundary), ‘telotron’ (≥10% coverage), ‘non-telotron’ (<1% coverage), with intermediate introns (1–10%) excluded from binary comparisons. The ‘strict de novo’ count (≥70% telomeric + degenerate) in the abstract uses a more inclusive threshold to capture degenerate-telomeric introns. All analyses used Python 3.10. Code is available at [repository URL upon publication].

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